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CA-SOFTWARE (CA-S)

ASSIGNMENT – 2

Principles in Practice - Draft a one-page scenario where you apply Microservices Architecture and Event-Driven Architecture to a hypothetical e-commerce platform. Outline how SOLID principles could enhance the design. Use bullet points to indicate how DRY and KISS principles can be observed in this context.

Scenario: E-commerce platform selling electronics with modules for orders, Inventory, Payments, shipping and Customer Notifications.

1 Microservices Architecture:

- Each module is developed as a separate microservice
- Services communicate over HTTP/REST or message brokers (e.g. RabbitMQ)
- Independent deployment and scaling

2 Event - Driven Architecture :

- Services Publish events (e.g., "orderPlaced", "Payment confirmed")
- Other services react to these events asynchronously.
- Enhances decoupling and responsiveness

3 Application of SOLID Principles:

- S: Each services has a single responsibility (e.g., Inventory only manages stock)
- O: service can be extended (e.g., add discounts) without modifying existing logic.
- L: Replace payment gateway implementation without changing client code.
- I: Services define only necessary interface (e.g., Notifications service has a simple send()).

- D : High-level services depends on abstractions, not on lower level modules directly.

4 DRY (Don't Repeat Yourself):

- Shared components like logging, authentication are abstracted into common services or libraries.

5 KISS (Keep it Simple, Stupid):

- Services focus on a specific task, simple REST APIs, avoid over-engineering.